

1. Stack levantado y todos los servicios accesibles

```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose up -d --build
#18 exporting manifest list sha256:de26f9021006e58ad3bfb039beec5def24e21e63ef733a591d883df11ada49b 0.0s done
#18 naming to docker.io/library/iac-observabilidad-backend:latest done
#18 unpacking to docker.io/library/iac-observabilidad-backend:latest done
#18 DONE 0.1s

#19 [backend] resolving provenance for metadata file
#19 DONE 0.0s

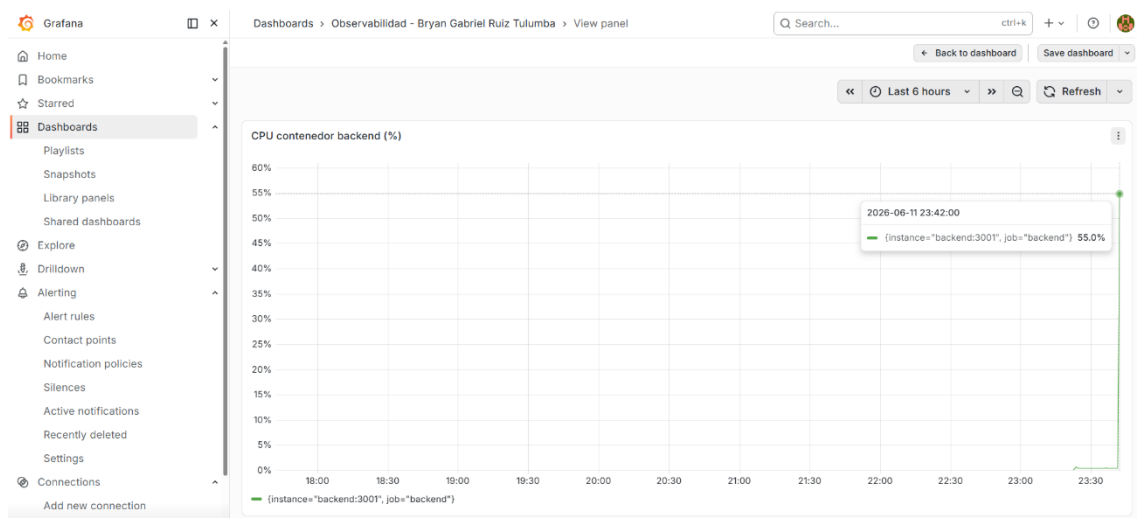
#20 [frontend] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 13/13
✓ Image iac-observabilidad-backend          Built
✓ Image iac-observabilidad-frontend         Built
✓ Network iac-observabilidad_default        Created
✓ Volume iac-observabilidad_prometheus-data Created
✓ Volume iac-observabilidad_grafana-data    Created
✓ Container lab-prometheus                  Started
✓ Container lab-cadvisor                    Started
✓ Container lab-loki                        Started
✓ Container lab-backend                     Started
✓ Container lab-node-exporter               Started
✓ Container lab-frontend                    Started
... 2 more
```

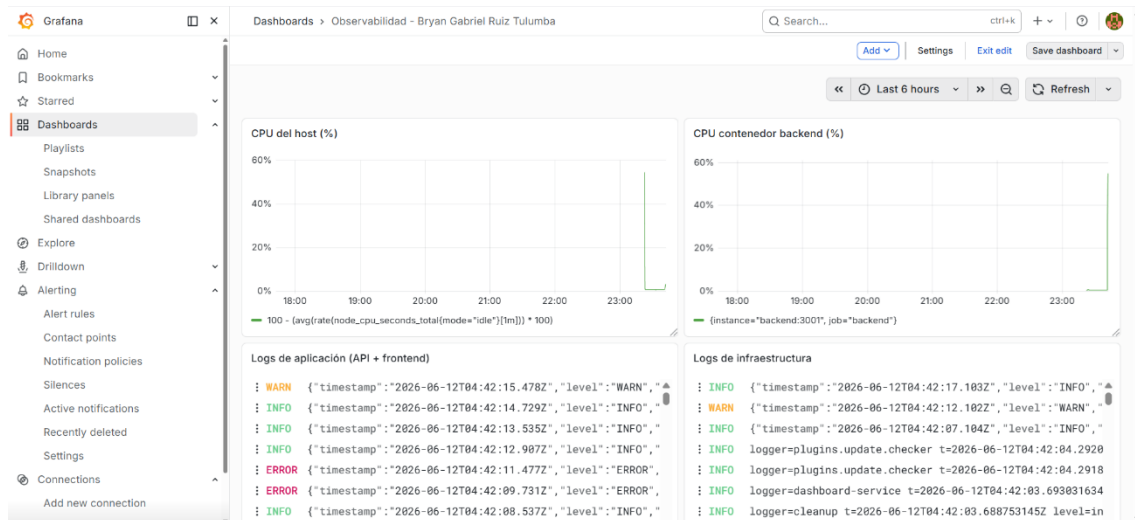
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
lab-alloy	grafana/alloy:v1.10.0	"/bin/alloy run --se..."	alloy	12 seconds ago	Up 10 seconds
lab-backend	iac-observabilidad-backend	"docker-entrypoint.s..."	backend	12 seconds ago	Up 11 seconds
lab-cadvisor	gcr.io/cadvisor/cadvisor:v0.49.1	"/usr/bin/cadvisor -..."	cadvisor	12 seconds ago	Up 11 seconds (h
lab-frontend	iac-observabilidad-frontend	"docker-entrypoint.s..."	frontend	12 seconds ago	Up 10 seconds
lab-grafana	grafana/grafana:12.4.0	"/run.sh"	grafana	12 seconds ago	Up 10 seconds
lab-loki	grafana/loki:3.4.0	"/usr/bin/loki --con..."	loki	12 seconds ago	Up 11 seconds
lab-node-exporter	prom/node-exporter:v1.8.2	"/bin/node_exporter ..."	node-exporter	12 seconds ago	Up 11 seconds
lab-prometheus	prom/prometheus:latest	"/bin/prometheus --c..."	prometheus	12 seconds ago	Up 11 seconds

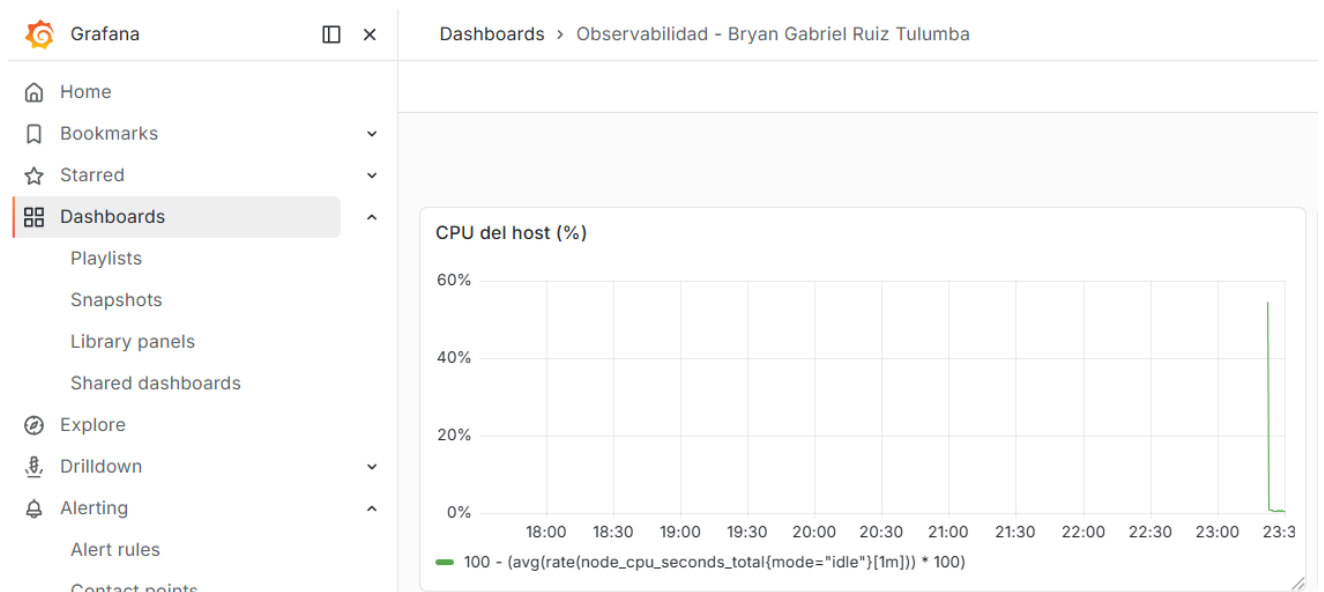
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad>
```

2. Dashboard con panel de CPU por contenedor (con umbral en 50%)





3. Dashboard con panel de CPU del host



4. Panel de logs de aplicación funcionando (filtro por nivel) INFO

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
INFO {"timestamp":"2026-06-12T05:41:15.376Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":7125,"amount":146}
INFO {"timestamp":"2026-06-12T05:41:14.579Z","level":"INFO","service":"frontend","msg":"click_boton","button":"cargar"}
INFO {"timestamp":"2026-06-12T05:41:13.284Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:12.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:11.378Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":1573,"amount":345}
INFO {"timestamp":"2026-06-12T05:41:09.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:41:08.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:07.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:03.378Z","level":"INFO","service":"backend","msg":"usuario_autenticado","user_id":5}
INFO {"timestamp":"2026-06-12T05:41:03.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:02.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:59.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:40:58.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:40:57.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:55.377Z","level":"INFO","service":"backend","msg":"pago_procesado","order_id":9885,"gateway":"st"}
```

WARN

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
WARN {"timestamp":"2026-06-12T05:41:04.588Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:54.581Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:40:24.582Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:20.638Z","level":"WARN","service":"frontend","msg":"carga_cpu_solicitada_desde_frontend"}
WARN {"timestamp":"2026-06-12T05:40:20.637Z","level":"WARN","service":"backend","msg":"cpu_load_test_started","seconds":30}
WARN {"timestamp":"2026-06-12T05:40:07.388Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5991,"attempt":1}
WARN {"timestamp":"2026-06-12T05:39:39.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:39:14.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:39:11.386Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":1425}
WARN {"timestamp":"2026-06-12T05:38:55.388Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":1534,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:43.379Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":2020,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:31.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5601,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:27.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":7988,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:19.584Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:38:15.382Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":2160}
```

CPU contenedor backend (%)

Logs de infraestructura

ERROR

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

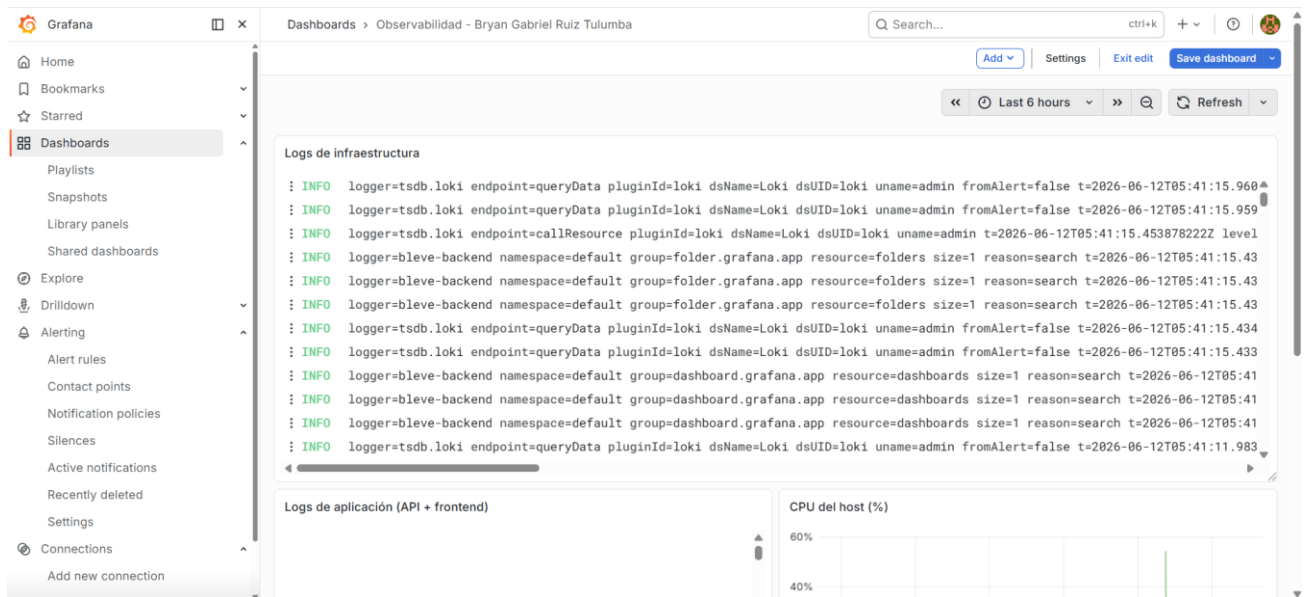
Logs de aplicación (API + frontend)

```
ERROR {"timestamp":"2026-06-12T05:41:07.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"c57586"}
ERROR {"timestamp":"2026-06-12T05:40:59.377Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:51.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"47a352"}
ERROR {"timestamp":"2026-06-12T05:40:44.580Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:27.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:19.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:19.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"ab1832"}
ERROR {"timestamp":"2026-06-12T05:40:14.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:11.380Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"b643bd"}
ERROR {"timestamp":"2026-06-12T05:40:03.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:39:59.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:39:59.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"f7faf6"}
ERROR {"timestamp":"2026-06-12T05:39:55.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"23d1f0"}
ERROR {"timestamp":"2026-06-12T05:39:51.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"fc54a3"}
ERROR {"timestamp":"2026-06-12T05:39:44.581Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
```

CPU contenedor backend (%)

Logs de infraestructura

5. Panel de logs de infraestructura funcionando



6. Alarma de CPU > 50% configurada correctamente

The screenshot shows the Grafana 'Alerting' configuration steps for a new alert rule. The steps are:

- 1. Enter alert rule name**
Enter a name to identify your alert rule.
Name: CPU backend > 50%
- 2. Define query and alert condition**
Define query and alert condition. [Need help?](#)
Prometheus [Options](#) 10m to now, MD = 43200, Min. Interval = 1s
Explain [Metrics browser](#) [Run queries](#) [Builder](#) [Code](#)
rate(backend_process_cpu_seconds_total[1m]) * 100
Options Legend: Auto Format: Time series Step: auto Type: Instant
Table
{instance="backend:3001", job="backend"} 0.46133
Alert condition
WHEN QUERY IS ABOVE 50
{instance="backend:3001", job="backend"} 0 Normal
[Preview alert rule condition](#)
- 3. Add folder and labels**
Organize your alert rule with a folder and set of labels. [Need help?](#)
Folder
Select a folder to store your rule in.
alerta [+ New folder](#)
Labels
Add labels to your rule for searching, silencing, or routing to a notification policy. [Need help?](#)
severity: warning [Edit labels](#)
- 4. Set evaluation behavior**
Define how the alert rule is evaluated. [Need help?](#)
Evaluation group and interval
alertas or [+ New evaluation group](#)
All rules in the selected group are evaluated every 10s.
Pending period
Period during which the threshold condition must be met to trigger an alert. Selecting "None" triggers the alert immediately once the condition is met.
30s
None 10s 20s 30s 40s 50s
Keep firing for
Period during which the alert will continue to show up as firing even though the threshold condition is no longer breached. Selecting "None" means the alert will be back to normal immediately.
0s
None 10s 20s 30s 40s 50s

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Alerting > Alert rules > New alert rule

Q Search...ctrl+k

5. Configure notifications

Select who should receive a notification when an alert rule fires.

Need help?

Recipient

Alertmanager: grafana

Contact point

empty

View or create contact points

Muting, grouping and timings (optional)

6. Configure notification message

Add more context to your alert notifications.

Need help?

Summary (optional)

Short summary of what happened and why.

Enter a summary...

Description (optional)

Description of what the alert rule does.

Enter a description...

Runbook URL (optional)

Webpage where you keep your runbook for the alert.

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Alerting > alerta > alertas > CPU backend > 50%

Q Search...ctrl+k

CPU backend > 50%Normal

Notifications are delivered toempty

Evaluation intervalEvery 10s

Labelsseveritywarning

Edit

More

Query and conditions

Instances

History

Details

Versions

A Prometheus10mto now

View in Explore

rate[backend_process_cpu_seconds_total[1m]] * 100

Table

{instance="backend:3001", job="backend"}0.49349

C Threshold

Alert condition

Input A Is above 50

{instance="backend:3001", job="backend"}0Normal

7. Evidencia de la alarma en estado *Firing* tras generar carga EN PENDING

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Q Search...ctrl+k

Alert rules

+ New alert rule

Export rule definition

Rules that determine whether an alert will fire

Search by data sources

All data sources

Dashboard

Select dashboard

State

Firing

Normal

Pending

Recovering

Rule type

Alert

Recording

Health

Ok

No Data

Error

Contact point

Choose a contact point

Search

Q Search

View as

Grouped

List

State

1 rule1 pending

Grafana-managed

Export rules

New recording rule

alerta > alertas

1 pending10s

State	Name	Health	Summary	Next evaluation	Actions
Pending	CPU backend > 50%	ok		in a few seconds	ViewEditMore

RESULTADO EN FIRING

The screenshot shows the Grafana Alerting 'Alert rules' page. On the left is a sidebar with navigation options like Home, Bookmarks, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Explore, Drilldown, Alerting, and Connections. The 'Alerting' section is expanded, showing 'Alert rules' as the selected item. The main panel displays 'Alert rules' with a search bar and filters for 'State' (Firing, Normal, Pending, Recovering), 'Rule type' (Alert, Recording), 'Health' (Ok, No Data, Error), and 'Contact point'. A search bar is also present. Below the filters, it shows '1 rule 1 firing'. The rule is 'Grafana-managed' and named 'alerta > alertas'. A table lists the alert rule with columns: State, Name, Health, Summary, Next evaluation, and Actions. The rule is in a 'Firing' state for 9s, with the summary 'CPU backend > 50%' and health 'ok'. The next evaluation is 'in a few seconds'. The actions are 'View', 'Edit', and 'More'. At the bottom, there is a search bar with the text 'Búsqueda - Praia da Ursa, Portugal'.

8. Ciclo cerrado alarma → log vía webhook

The screenshot shows a Grafana dashboard titled 'Dashboards > Observabilidad - Bryan Gabriel Ruiz Tulumba > Edit panel'. The dashboard is in 'Table view' and shows 'Logs de aplicación (API + frontend)'. The logs are displayed in a table with columns: timestamp, level, service, msg, and alert_status. The logs show three error messages: 'grafana_alert_received', 'grafana_alert_received', and 'grafana_alert_received'. The dashboard also has a 'Queries' section with a query for 'application' and a 'Transformations' section. The query is '{tier="application"} | json | msg =~ ".*alert.*"'. The dashboard has a search bar and a 'Refresh' button.

Breve explicación (5–8 líneas) de qué hace cada componente del stack.

Prometheus es el motor de las métricas del stack. Hace scraping periódico a los endpoints de cada servicio y los almacena. Permite consultarlas con PromQL.

Node Exporter expone las métricas del host (CPU, memoria, disco) en formato Prometheus. Y necesita acceso al sistema de archivos del host

cAdvisor hace lo mismo que node-exporter pero permite ver cuántos recursos consume cada contenedor Docker por separado.

Loki es como Prometheus pero para logs de cada contenedor. Indexa solo sus etiquetas lo que lo hace muy liviano

Grafana Alloy es el recolector de logs, captura el stdout de cada contenedor y los reenvía a Loki con etiquetas como tier="application" o tier="infrastructure".

Grafana se conecta a Prometheus y Loki como fuentes de datos y permite construir dashboards con métricas y logs. También gestiona las alarmas.

Backend y Frontend son las aplicaciones que usamos en el lab. Exponen métricas propias y emiten logs en formato JSON, siendo los sujetos de observación del stack.

LINK PROYECTO DE IAC:

https://github.com/Estalin7/proyecto_infra